1.

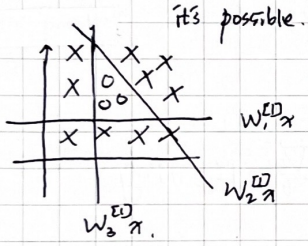
(a) $\frac{\partial J}{\partial W_{1,2}^{[1]}} = \frac{\partial J}{\partial O} \cdot \frac{\partial O}{\partial z^{[2]}} \cdot \frac{\partial z^{[2]}}{\partial a^{[1]}} \cdot \frac{\partial a^{[1]}}{\partial z^{[1]}} \cdot \frac{\partial z^{[1]}}{\partial W_{1,2}^{[1]}}$

※ $a^{[1]} = \sigma(W^{[1]}x)$, $O = a^{[2]} = \sigma(W^{[2]}a^{[1]})$

$$= \frac{1}{m}\sum_{i=1}^{m} 2(O^{(i)} - y^{(i)}) \cdot O^{(i)}(1-O^{(i)}) \cdot W_2^{[2]} \cdot a^{[1](i)}(1-a^{[1](i)}) x_1^{(i)}$$

$$\therefore W_{1,2}^{[1]} := W_{1,2}^{[1]} - \alpha \cdot \frac{2}{m} W_2^{[2]} \sum_{i=1}^{m} (O^{(i)} - y^{(i)}) O^{(i)}(1-O^{(i)}) a^{[1](i)}(1-a^{[1](i)}) x_1^{(i)}$$

(b) it's possible.



(c) it isn't possible. it is just linear regression.

2.

(a)

$$D_{KL}(p\|Q) = \sum_x p(x) \log \frac{p(x)}{Q(x)} = -\sum_x p(x) \log \frac{Q(x)}{p(x)}$$

$$= E_{x\sim p(x)}\left[-\log \frac{Q(x)}{p(x)}\right] \geq -\log E_{x\sim p(x)}\left[\frac{Q(x)}{p(x)}\right]$$

$$= -\log \sum_x p(x) \frac{Q(x)}{p(x)} = -\log 1$$

$$= 0.$$

$p = Q \Rightarrow \sum_x p(x) \log 1 = 0.$

$D_{KL}(p\|Q) = 0 \Rightarrow \frac{Q(x)}{p(x)} = E\left[\frac{Q(x)}{p(x)}\right] = 1 \therefore p = Q.$

(b)

$$D_{KL}(p(x)\|Q(x)) + D_{KL}(p(Y|X)\|Q(Y|X))$$

$$= \sum_x p(x) \log \frac{p(x)}{Q(x)} + \sum_x p(x) \left(\sum_y p(y|x) \log \frac{p(y|x)}{Q(y|x)}\right)$$

$$= \sum_x p(x) \log \frac{p(x)}{Q(x)} + \sum_x p(x) \sum_y \frac{p(x,y)}{p(x)} \log \frac{p(x,y)/p(x)}{Q(x,y)/Q(x)}$$

$$= \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{Q(x,y)}$$

$$= D_{KL}(p(X,Y)\|Q(X,Y))$$

motemote

(c)

$$D_{KL}(\hat{p} \| p_\theta) = \sum_x \hat{p}(x) \log \frac{\hat{p}(x)}{p_\theta(x)} = \sum_x \hat{p}(x) \log \hat{p}(x) - \sum_x \hat{p}(x) \log p_\theta(x)$$

$$\begin{aligned}
\arg\min_\theta D_{KL}(\hat{p} \| p_\theta) &= \arg\min_\theta \left(-\sum_x \hat{p}(x) \log p_\theta(x)\right) \\
&= \arg\max_\theta \left(\sum_x \hat{p}(x) \log p_\theta(x)\right) \\
&= \arg\max_\theta \left(\sum_x \left(\frac{1}{m} \sum_{i=1}^m 1\{x^{(i)} = x\}\right) \log p_\theta(x)\right) \\
&= \arg\max_\theta \sum_{i=1}^m \log p_\theta(x^{(i)})
\end{aligned}$$

3 (a)

$$\begin{aligned}
E_{y \sim p(y;\theta)}\left[\nabla_{\theta'} \log p(y;\theta')\big|_{\theta'=\theta}\right] &= \int_{-\infty}^{\infty} p(y;\theta) \nabla_\theta \log p(y;\theta)\, dy \\
&= \int_{-\infty}^{\infty} p(y;\theta) \frac{\nabla_\theta p(y;\theta)}{p(y;\theta)} dy \\
&= \nabla_\theta \int_{-\infty}^{\infty} p(y;\theta)\, dy \\
&= \nabla_\theta 1 = 0.
\end{aligned}$$

(b)

$$\begin{aligned}
I(\theta) &= \mathrm{Cov}_{y \sim p(y;\theta)}\left[\nabla_{\theta'} \log p(y;\theta')\big|_{\theta'=\theta}\right] \\
&= E_{y \sim p(y;\theta)}\left[\nabla_{\theta'} \log p(y;\theta') \nabla_{\theta'} \log p(y;\theta')^T\big|_{\theta'=\theta}\right]
\end{aligned}$$

$$\left(\because E\left[\nabla_\theta \log p(y;\theta)\big|_{\theta'=\theta}\right] = 0.\right.$$

$$\left.\mathrm{Cov}[X] = E[(X - E[X])(X - E[X])^T]\right)$$

(c)

$$E_{y\sim p(y;\theta)}\left[\nabla_\theta \log p(y;\theta)\big|_{\theta'=\theta}\right]=0, \quad \therefore \nabla_\theta E[\cdot]=0.$$

$$\nabla_\theta \int_{-\infty}^{\infty} p(y;\theta)\,\nabla_\theta \log p(y;\theta)\,dy = 0.$$

$$\int_{-\infty}^{\infty}\left(\nabla_\theta p(y;\theta)\,\nabla_\theta \log p(y;\theta) + p(y;\theta)\,\nabla_\theta^2 \log p(y;\theta)\right)dy = 0.$$

$$\int_{-\infty}^{\infty} \nabla_\theta p(y;\theta)\,\nabla_\theta \log p(y;\theta)\,dy = -\int_{-\infty}^{\infty} p(y;\theta)\,\nabla_\theta^2 \log p(y;\theta)\,dy$$

$$\int_{-\infty}^{\infty} p(y;\theta)\,\nabla_\theta \log p(y;\theta)\,\nabla_\theta \log p(y;\theta)\,dy \qquad E\left[-\nabla_\theta^2 \log p(y;\theta)\right]$$

$$\left(\because \nabla_\theta \log p(y;\theta) = \frac{1}{p(y;\theta)}\nabla_\theta p(y;\theta)\right)$$

$$\left(\therefore \nabla_\theta p(y;\theta) = p(y;\theta)\,\nabla_\theta \log p(y;\theta)\right)$$

$$E\left[\nabla_\theta \log p(y;\theta)\,\nabla_\theta \log p(y;\theta)^T\right]$$

$$I(\theta)$$

$$\therefore I(\theta) = E_{y\sim p(y;\theta)}\left[-\nabla_\theta^2 \log p(y;\theta)\big|_{\theta'=\theta}\right]$$

motemote

(d) $\tilde{\theta} = \theta + d$.

$f(\tilde{\theta}) = D_{KL}(P_\theta \| P_{\tilde{\theta}})$

$f(\tilde{\theta}) \approx D_{KL}(P_\theta \| P_\theta) + (\tilde{\theta}-\theta)^T \nabla_{\theta'} f(\theta')|_{\theta'=\theta} + \frac{1}{2}(\tilde{\theta}-\theta)^T (\nabla_{\theta'}^2 f(\theta')|_{\theta'=\theta})(\tilde{\theta}-\theta)$

$= 0 + d^T \nabla_{\theta'} D_{KL}(P_\theta \| P_{\theta'})|_{\theta'=\theta} + \frac{1}{2} d^T \nabla_{\theta'}^2 D_{KL}(P_\theta \| P_{\theta'})_{\theta'=\theta} d$

$\nabla_{\theta'} D_{KL}(P_\theta \| P_{\theta'})|_{\theta'=\theta} = \nabla_{\theta'} \int P_\theta(x) \log \frac{P_\theta(x)}{P_{\theta'}(x)} dx$

$= \nabla_{\theta'} \int P_\theta(x) (\log P_\theta(x) - \log P_{\theta'}(x)) dx.$

$= -\int P_\theta(x) \nabla_{\theta'} \log P_{\theta'}(x) dx.$

$= -E[\nabla_{\theta'} \log P_{\theta'}(x)|_{\theta'=\theta}] = 0$ (by (a))

$\nabla_{\theta'}^2 D_{KL}(P_\theta \| P_{\theta'})|_{\theta'=\theta} = \nabla_{\theta'}^2 \int P_\theta(x) \log \frac{P_\theta(x)}{P_{\theta'}(x)} dx$

$= \int P_\theta(x) \nabla_{\theta'}^2 (-\log P_{\theta'}(x)) dx.$

$= E[-\nabla_{\theta'}^2 \log P_{\theta'}(x)|_{\theta'=\theta}] = I(\theta)$ (by (c))

$\therefore f(\tilde{\theta}) = D_{KL}(P_\theta \| P_{\theta+d}) \approx \frac{1}{2} d^T I(\theta) d$

(e)

$$d^* = \arg\max_d \ell(\theta+d)$$

s.t

$$D_{KL}(P_\theta \| P_{\theta+d}) = c$$

$$\ell(\theta+d) \approx \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

$$D_{KL}(P_\theta \| P_{\theta+d}) \approx \frac{1}{2} d^T \mathcal{I}(\theta) d$$

$$\mathcal{L}(d, \lambda) = \ell(\theta+d) - \lambda (D_{KL}(P_\theta \| P_{\theta+d}) - c)$$

$$\approx \ell(\theta) + d^T \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \left(\frac{1}{2} d^T \mathcal{I}(\theta) d - c\right)$$

$$\nabla_d \mathcal{L}(d, \lambda) \approx \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta} - \lambda \mathcal{I}(\theta) d = 0$$

$$\therefore \tilde{d} = \frac{1}{\lambda} \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

$$\nabla_\lambda \mathcal{L}(d, \lambda) \approx c - \frac{1}{2} d^T \mathcal{I}(\theta) d = 0$$

$$= c - \frac{1}{2} \left(\frac{1}{\lambda} \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}\right)^T \mathcal{I}(\theta) \left(\frac{1}{\lambda} \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}\right)$$

$$= c - \frac{1}{2\lambda^2} \nabla_{\theta'} \ell(\theta')^T \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')$$

$$= 0.$$

$$\therefore \lambda = \sqrt{\frac{1}{2c} \nabla_{\theta'} \ell(\theta')^T \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')}$$

(f) Natural gradient :

$$\theta := \theta + \tilde{d}$$

$$= \theta + \frac{1}{\lambda} \mathcal{I}(\theta)^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

$$= \theta - \frac{1}{\lambda} E[\nabla^2_{\theta'} \ell(\theta')|_{\theta'=\theta}]^{-1} \nabla_{\theta'} \ell(\theta')|_{\theta'=\theta}$$

$$= \theta - \frac{1}{\lambda} E[H]^{-1} \nabla_\theta \ell(\theta)$$

Newton's Method : $\theta := \theta - H^{-1} \nabla_\theta \ell(\theta)$

motemote

4

(a)

$$\ell(\theta^{(t)}) = \ell_u(\theta^{(t)}) + \alpha \ell_s(\theta^{(t)})$$

$$= \sum_i^m \left(\sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t)})}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \left(\sum_i^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \theta^{(t)}) \right)$$

$$\leq \sum_i^m \left(\sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t+1)})}{Q_i^{(t)}(z^{(i)})} \right) + \alpha \sum_i^{\tilde{m}} \log p(\tilde{x}^{(i)}, \tilde{z}^{(i)}; \theta^{(t+1)})$$

$$\leq \ell_u(\theta^{(t+1)}) + \alpha \ell_s(\theta^{(t+1)}) = \ell(\theta^{(t+1)})$$

(b)

$$w_j^{(i)} = p(z^{(i)} = j \mid x^{(i)}; \theta)$$

$$= \frac{p(x^{(i)} \mid z^{(i)} = j; \mu_j, \Sigma_j)\, p(z^{(i)} = j; \phi_j)}{\sum_{l=1}^k p(x^{(i)} \mid z^{(i)} = l; \mu_l, \Sigma_l)\, p(z^{(i)} = l; \phi_l)}$$

$$= \frac{\mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)\phi_j}{\sum_{l=1}^k \mathcal{N}(x^{(i)}; \mu_l, \Sigma_l)\phi_l}$$

(c)

$$\phi_j = \frac{1}{m + \alpha\tilde{m}} \left(\sum_{i=1}^m w_j^{(i)} + \alpha \sum_{i=1}^{\tilde{m}} 1\{\tilde{z}^{(i)} = j\} \right)$$

$$\mu_j = \left(\sum_{i=1}^m w_j^{(i)} x^{(i)} + \alpha \sum_{i=1}^{\tilde{m}} 1\{\tilde{z}^{(i)} = j\} \tilde{x}^{(i)} \right) \Big/ \left(\sum_{i=1}^m w_j^{(i)} + \alpha \sum_{i=1}^{\tilde{m}} 1\{\tilde{z}^{(i)} = j\} \right)$$

$$\Sigma_j = \frac{\sum_{i=1}^m w_j^{(i)} (x^{(i)} - \mu_j)(x^{(i)} - \mu_j)^T + \alpha \sum_{i=1}^{\tilde{m}} 1\{\tilde{z}^{(i)} = j\} (\tilde{x}^{(i)} - \mu_j)(\tilde{x}^{(i)} - \mu_j)^T}{\sum_{i=1}^m w_j^{(i)} + \alpha \sum_{i=1}^{\tilde{m}} 1\{\tilde{z}^{(i)} = j\}}$$

(f) Semi-supervised EM take much less iterations, are more stable, higher quality.

5

(b)

original : $3 \times 8 = 24$ bits per pixel

compressed : 4 bits (16 colors) per pixel

$\Rightarrow 1/6$ 로 압축!